

# Legenda parametri

| Parametro | Significato                      |
|-----------|----------------------------------|
| k         | numero di letture dell'annealer  |
| TIMES     | quante volte si esegue il codice |
| $\lambda$ | $\lambda = (\sum p_i) / 3$       |

# Dati sperimentali - KSP 3

Parametri: k = 1000, TIMES = 5

## Risultato DP

| Profit | Weight |
|--------|--------|
| 2679   | 496    |

## Risultati QALS

| i_max | N_max | d_min | Avg fQ        | fQ Min Found  | Profit Found | Weight Found | Profit GAP | Weight GAP | n Min |
|-------|-------|-------|---------------|---------------|--------------|--------------|------------|------------|-------|
| 10    | 5     | 4     | 1699878079.43 | 1199097503.77 | 4101         | 2614         | -1422      | -2118      | 1     |
| 50    | 25    | 18    | 849939039.71  | 925804052.7   | 3030         | 2373         | -351       | -1877      | 1     |
| 100   | 50    | 35    | 566626026.48  | 445660422.37  | 2880         | 1849         | -201       | -1353      | 1     |
| 250   | 125   | 88    | 424969519.86  | 217137783.97  | 2419         | 1508         | 260        | -1012      | 1     |
| 500   | 250   | 175   | 339975615.89  | 217137783.97  | 2419         | 1508         | 260        | -1012      | 1     |
| 1000  | 500   | 350   | 283313013.24  | 217137783.97  | 2419         | 1508         | 260        | -1012      | 1     |

## Risultati NO QALS

| Avg fQ       | fQ Min Found | Profit Found | Weight Found | Profit GAP | Weight GAP | n Min |
|--------------|--------------|--------------|--------------|------------|------------|-------|
| -71426754.12 | -71426766.2  | 2483         | 501          | 196        | -5         | 3     |

# Dati sperimentali - KSP 3

Parametri: k = 1250, TIMES = 5

## Risultato DP

| Profit | Weight |
|--------|--------|
| 2679   | 496    |

## Risultati QALS

| i_max | N_max | d_min | Avg fQ        | fQ Min Found | Profit Found | Weight Found | Profit GAP | Weight GAP | n Min |
|-------|-------|-------|---------------|--------------|--------------|--------------|------------|------------|-------|
| 10    | 5     | 4     | 1995599883.65 | 1490744151.9 | 4473         | 2844         | -1794      | -2348      | 1     |
| 50    | 25    | 18    | 997799941.83  | 354904183.2  | 2493         | 1725         | 186        | -1229      | 1     |
| 100   | 50    | 35    | 665199961.22  | 262954985.97 | 2652         | 1585         | 27         | -1089      | 1     |
| 250   | 125   | 88    | 498899970.91  | 262954985.97 | 2652         | 1585         | 27         | -1089      | 1     |
| 500   | 250   | 175   | 399119976.73  | 262954985.97 | 2652         | 1585         | 27         | -1089      | 1     |
| 1000  | 500   | 350   | 332599980.61  | 262954985.97 | 2652         | 1585         | 27         | -1089      | 1     |

## Risultati NO QALS

| Avg fQ      | fQ Min Found | Profit Found | Weight Found | Profit GAP | Weight GAP | n Min |
|-------------|--------------|--------------|--------------|------------|------------|-------|
| -71426749.6 | -71426766.2  | 2483         | 501          | 196        | -5         | 3     |

# Dati sperimentali - KSP 3

Parametri: k = 1500, TIMES = 5

## Risultato DP

| Profit | Weight |
|--------|--------|
| 2679   | 496    |

## Risultati QALS

| i_max | N_max | d_min | Avg fQ        | fQ Min Found  | Profit Found | Weight Found | Profit GAP | Weight GAP | n Min |
|-------|-------|-------|---------------|---------------|--------------|--------------|------------|------------|-------|
| 10    | 5     | 4     | 1890742732.08 | 1585482114.47 | 4194         | 2914         | -1515      | -2418      | 1     |
| 50    | 25    | 18    | 945371366.04  | 872207870.6   | 3454         | 2322         | -775       | -1826      | 1     |
| 100   | 50    | 35    | 630247577.36  | 526874638.27  | 2605         | 1951         | 74         | -1455      | 1     |
| 250   | 125   | 88    | 472685683.02  | 422150941.4   | 2938         | 1818         | -259       | -1322      | 1     |
| 500   | 250   | 175   | 378148546.42  | 422150941.4   | 2938         | 1818         | -259       | -1322      | 1     |
| 1000  | 500   | 350   | 315123788.68  | 422150941.4   | 2938         | 1818         | -259       | -1322      | 1     |

## Risultati NO QALS

| Avg fQ      | fQ Min Found | Profit Found | Weight Found | Profit GAP | Weight GAP | n Min |
|-------------|--------------|--------------|--------------|------------|------------|-------|
| -71426766.2 | -71426766.2  | 2483         | 501          | 196        | -5         | 5     |

## Dati sperimentali - KSP 3

Parametri: k = 1750, TIMES = 5

### Risultato DP

| Profit | Weight |
|--------|--------|
| 2679   | 496    |

### Risultati QALS

| i_max | N_max | d_min | Avg fQ        | fQ Min Found  | Profit Found | Weight Found | Profit GAP | Weight GAP | n Min |
|-------|-------|-------|---------------|---------------|--------------|--------------|------------|------------|-------|
| 10    | 5     | 4     | 2101877204.07 | 1347504684.27 | 4376         | 2734         | -1697      | -2238      | 1     |
| 50    | 25    | 18    | 1050938602.03 | 670787159.27  | 2170         | 2116         | 509        | -1620      | 1     |
| 100   | 50    | 35    | 700625734.69  | 458008387.87  | 2393         | 1865         | 286        | -1369      | 1     |
| 250   | 125   | 88    | 525469301.02  | 172055577.57  | 2587         | 1426         | 92         | -930       | 1     |
| 500   | 250   | 175   | 420375440.81  | 172055577.57  | 2587         | 1426         | 92         | -930       | 1     |
| 1000  | 500   | 350   | 350312867.34  | 172055577.57  | 2587         | 1426         | 92         | -930       | 1     |

### Risultati NO QALS

| Avg fQ      | fQ Min Found | Profit Found | Weight Found | Profit GAP | Weight GAP | n Min |
|-------------|--------------|--------------|--------------|------------|------------|-------|
| -71426766.2 | -71426766.2  | 2483         | 501          | 196        | -5         | 5     |

# Dati sperimentali - KSP 3

Parametri: k = 2000, TIMES = 5

## Risultato DP

| Profit | Weight |
|--------|--------|
| 2679   | 496    |

## Risultati QALS

| i_max | N_max | d_min | Avg fQ        | fQ Min Found | Profit Found | Weight Found | Profit GAP | Weight GAP | n Min |
|-------|-------|-------|---------------|--------------|--------------|--------------|------------|------------|-------|
| 10    | 5     | 4     | 1793249081.29 | 1234211629.1 | 3886         | 2643         | -1207      | -2147      | 1     |
| 50    | 25    | 18    | 896624540.65  | 92938955.77  | 2330         | 1261         | 349        | -765       | 1     |
| 100   | 50    | 35    | 597749693.76  | 92938955.77  | 2330         | 1261         | 349        | -765       | 1     |
| 250   | 125   | 88    | 448312270.32  | 92938955.77  | 2330         | 1261         | 349        | -765       | 1     |
| 500   | 250   | 175   | 358649816.26  | 92938955.77  | 2330         | 1261         | 349        | -765       | 1     |
| 1000  | 500   | 350   | 298874846.88  | 92938955.77  | 2330         | 1261         | 349        | -765       | 1     |

## Risultati NO QALS

| Avg fQ      | fQ Min Found | Profit Found | Weight Found | Profit GAP | Weight GAP | n Min |
|-------------|--------------|--------------|--------------|------------|------------|-------|
| -71426766.2 | -71426766.2  | 2483         | 501          | 196        | -5         | 5     |